ABDUL JAMEEL AHAMED NAJAH

AI Engineer | LLM, RAG, Agentic Systems & MLOps

Sri Lanka (Open to Relocation | Available Immediately) | +94 741772234 | najahaja00@gmail.com

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PROFESSIONAL SUMMARY

AI Engineer and Computer Engineering graduate (Gold Medalist, Rank #1 in Batch, CGPA 3.94/4.00) specializing in LLM engineering, RAG, agentic systems, and production MLOps. Completed 3 AI and software engineering internships building real-time voice agents handling 200+ daily calls, fine-tuning and quantizing open-weights LLMs (QLoRA/AWQ) to 71.69% exact match, and deploying FastAPI microservices with automated drift detection and CI/CD pipelines. Based in Sri Lanka and available immediately.

TECHNICAL SKILLS

- **Generative AI & LLMs:** Fine-Tuning (QLoRA, LoRA, PEFT), AWQ/GPTQ Quantization, vLLM (PagedAttention), RAG Architectures, Agentic Orchestration, Hugging Face Transformers, LangChain, LangGraph, LangSmith, Prompt Engineering, LLM Benchmarking & Evaluation
- **Machine Learning & MLOps:** Python, Scikit-learn, PyTorch, TensorFlow, MLflow, Drift Detection (PSI, Kolmogorov–Smirnov), Champion/Challenger Gating, Automated Retraining, Reinforcement Learning (Stable-Baselines3), Computer Vision (YOLO11, SAM2), Model Evaluation
- **Backend, APIs & Databases:** FastAPI, Pydantic, SQLAlchemy, Node.js, REST APIs, Microservices, PostgreSQL (Neon), MySQL, SQLite, MongoDB, Vector Databases (FAISS, ChromaDB), SQL, JWT Authentication
- **Voice & Real-Time AI:** LiveKit, Retell AI, Dograh, WebRTC / Streaming Audio Pipelines
- **Testing, CI/CD & Tools:** Docker, GitHub Actions (CI/CD), Pytest, Ruff, Uvicorn, React.js, JavaScript, C++, Streamlit, Git, Linux/Bash

PROFESSIONAL EXPERIENCE

Artificizen | Lahore, Pakistan (Onsite)
AI Intern | *Jan 2026 - Jul 2026*

- Delivered an end-to-end real-time AI voice agent for a client project handling 200+ calls per day by orchestrating Gemini and Grok LLMs with LangChain and LangGraph on Retell, Dograh (Docker), and LiveKit.
- Achieved sub-10ms retrieval latency during live audio streams by engineering FastAPI microservices on MySQL and PostgreSQL (Neon) with FAISS and ChromaDB vector search.
- Decreased failed workflow runs by 5% through systematic testing and validation of multi-agent graph workflows across client projects.
- Aligned client stakeholders on system design by authoring end-to-end technical documentation and architecture blueprints.

Pinnacloid | Lahore, Pakistan (Onsite)
Full-Stack Developer Intern | *Aug 2025 - Oct 2025*

- Reduced endpoint request latency by 25% by optimizing 12+ REST API routes and retrieval queries supporting real-time RAG workflows.
- Shipped production-ready LLM-integrated features utilizing Node.js, Express, React.js, and MongoDB (MERN architecture).
- Diagnosed and resolved 30+ critical production bugs and led Node.js backend orchestration for cross-team feature integrations.

Arch Technologies | Remote
Machine Learning Intern | *Jun 2025 - Jul 2025*

- Adapted Llama-3.2-3B to domain-specific healthcare requirements using prompt engineering and medical Chain-of-Thought (CoT) datasets.
- Trained and evaluated a YOLO11 brain tumor detection pipeline on a 5-class Roboflow dataset, achieving 87.2% mAP@0.5 on held-out test data.

- Developed agentic workflows for automated multimodal medical data processing, integrating vector-database retrieval and voice-cloning services.

TECHNICAL PROJECTS

FT-Bench: LLM Fine-Tuning & Quantization Benchmark | Llama-3.2-3B, QLoRA, AWQ, vLLM, FastAPI, GitHub Actions

- Raised exact match from 2.62% to 71.69% and Slot F1 from 16.41% to 85.34% on 2,974 held-out samples by fine-tuning Llama-3.2-3B-Instruct with QLoRA (rank-16, 4-bit NF4) on 11,481 MASSIVE NLU samples (bootstrap CIs, n=1,000).
- Cut VRAM consumption by 39% (6.2 GB to 3.8 GB) while preserving 72.19% exact match via AWQ INT4 quantization (group size 128); modeled self-hosted serving at ~93% lower cost than closed-source APIs at 100K queries/day.
- Productionized inference and evaluation pipeline with FastAPI, vLLM (PagedAttention), 4-quadrant error analysis, 34 automated unit tests, GitHub Actions CI/CD, and published 3 Hugging Face model checkpoints.

ML-Ops-Live: End-to-End NYC Taxi Trip-Duration Prediction & MLOps Platform | Python, Scikit-learn, MLflow, FastAPI, SQLAlchemy, Pytest

- Engineered a Python, scikit-learn, and MLflow trip-duration prediction pipeline on 102,189 cleaned NYC TLC records, where HistGradientBoostingRegressor reduced offline holdout MAE by 14.1% (2.75 vs. 3.20 min) over a Ridge baseline and achieved 2.77 min out-of-time MAE ($R^2 = 0.755$).
- Built a FastAPI, Pydantic, and SQLAlchemy inference and monitoring service with SQLite request/audit logging, delayed ground-truth feedback (POST /feedback) for realized MAE/RMSE tracking, and PSI/Kolmogorov–Smirnov drift detection with 7-day calendar coverage validation.
- Automated drift-triggered retraining, champion/challenger gating, transactional model promotion/rollback, and CI (pytest, ruff) across 55 tests; verified a 38.6% MAE reduction in a controlled drift simulation and 292.4 ms P95 client latency (24.7 req/s) under local Uvicorn HTTP load testing.

AI Travel Planner Agent & SaaS Platform | LangGraph, Groq, Tavily, FastAPI, RAG, JWT

- Resolved 10+ dynamic user constraints across multi-turn itineraries by building a LangGraph agent with Groq LLMs, Tavily web search, human-in-the-loop plan verification, and LangSmith telemetry tracing.
- Delivered a modular FastAPI SaaS backend featuring RAG retrieval, real-time weather and exchange rate API integrations, and JWT authentication with role-based access control (RBAC).

Multi-Agent Disaster Response Simulation | Deep RL, Stable-Baselines3, OpenStreetMap, Streamlit | 1st Place, Department Open House

- Reduced simulated emergency response time by 15-20% versus uncoordinated baseline routing by engineering a multi-agent deep RL system on 50+ node OpenStreetMap road networks, equipped with interactive Streamlit and Plotly dashboards.

Semantic Web & Document RAG System | LangGraph, FAISS, Hugging Face Transformers

- Built a document RAG pipeline with FAISS and all-MiniLM-L6-v2 embeddings for contextual question answering using a LangGraph rerank, generate, and evaluate workflow.
- Implemented an automated evaluation harness scoring answers on context relevance, generation faithfulness, and semantic similarity.

EDUCATION

The University of Lahore | Lahore, Pakistan
Bachelor of Science in Computer Engineering | *Sep 2022 - Jul 2026*

- Gold Medalist & Department Rank #1 (Highest CGPA in Batch: 3.94 / 4.00) | Recipient of Fully Funded Allama Iqbal Scholarship

CERTIFICATIONS

- **Coursera / DeepLearning.AI:** Deep Learning Specialization; Machine Learning Specialization; Advanced React
- **freeCodeCamp:** Data Analysis & Scientific Computing with Python; JavaScript Algorithms & Data Structures

LANGUAGES

English (Proficient) | Tamil (Native) | Sinhala (Intermediate) | Urdu (Intermediate) | Arabic (Basic)